

Aditya Pravar Dhavala

daprar5@tamu.edu · (512) - 817 - 9719 · Dallas, Texas · LinkedIn: apdhavala

EDUCATION

Texas A&M University

Expected 05/2029

B.S. Mechanical Engineering

GPA: 4.0

Relevant Coursework: Linear Algebra, Multivariable Calculus, Physics Mechanics

EXPERIENCE

ACE Composites in Automotive Bootcamp

- Produced hand layups and vacuum infusions of carbon fiber
- Thermoformed plastics using molds
- Learned how to safely handle and dispose of hazardous ingredients

ASME Go-Kart Project

09/2025-05/2026

- Designed and 3D modeled parts and performed finite element analysis to identify material deformation
- Sourced performance components to improve safety and durability of go-kart
- Welded, machined, and CNC laser cut parts during go-kart assembly to ensure proper fit and reduce failure points

ACE Manufacturing Training

- Learned how to program CNC toolpaths for milling operations using CAM software
- Machined aluminum, steel, and plastics to accurate tolerances
- Assembled air piston out of machined parts

10/2025

UT High School Research Academy

06/2024

Computational Materials

- Wrote Python optimization algorithms to determine stable forms of atomic compounds
 - Analyzed VASP vector outputs to determine changes in volume
 - Simulated dopants on Lithium-harvesting compounds to prolong lifespan through repeated cycles
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SKILLS

Programming: Python, Java

CAD: Solidworks, Autodesk Fusion

Manufacturing: CNC Milling, Soldering, Welding, CAM toolpath generation, Composites fabrication

EXTRACURRICULARS

Member, American Society of Mechanical Engineers

08/2025 - Present

Texas A&M University